

AE-371: Radiation of [Ru(CO)(OH)₂(PNP{tBu})] (AE-310) 365 nm and white, 0.1 mM KOH

Date: 2024-10-30

Tags: Dihydroxo Radiation PNP
P(tBu)N(py)P(tBu) [Ru(CO)(OH)₂(PNP)]
NMR python AE 1H 31P

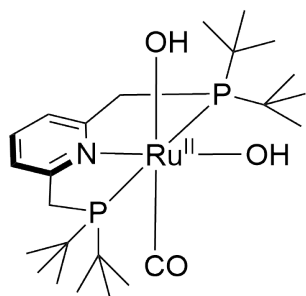
Category: Two-photon

Status: Done

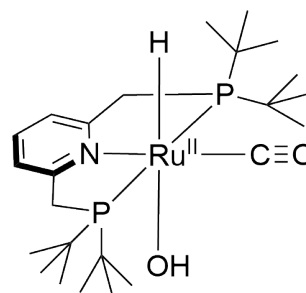
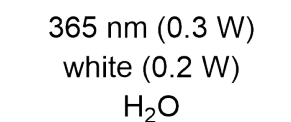
Created by: Alexander Eith

Reaction scheme/sample structure

Radiation of Dihydroxo



Chemical Formula: C₂₄H₄₅NO₃P₂Ru
Molecular Weight: 558,64552
[Ru(CO)(OH)₂(PNP^{tBu})]



Chemical Formula: C₂₄H₄₅NO₂P₂Ru
Molecular Weight: 542,64652
[Ru(CO)H(OH)(PNP^{tBu})]

Literature/reference experiments

Literature	Procedure after https://doi.org/10.1039/D1EE01053K
Reproduction	/
Related experiment	Two-photon - AE-357: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, 1000 mM KOH Two-photon - AE-358: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, 50 mM KOH Two-photon - AE-362: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, 5 mM KOH I Two-photon - AE-363: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, 1 mM KOH

Reagents

Name	CAS/Experiment number	Amount [μmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume [ml]
[Ru(CO)(OH) ₂ (PNP{tBu})]	AE-310: Synthesis of [Ru(CO)(OH) ₂ (PNP ^{tBu})] using Ag ₂ O, 5 h (AE-310-1)	4.48	/	2.50	2.54	558.65	/

KOH (5 mM)	Prep work - AE-361: 5 mM Stocksolution of KOH in water	/	/	/	/	/	0.012
Milli-Q water	KRA-104: Degassing of Milli-Q water	/	/	/	/	/	0.88

Irradiation Parameters

Power measured using [Power Meter] 843-R-USB + 919P-020-12 unless specified otherwise.

	Light Source Name	Wavelength [nm]	Power Setting [mW]
First light source	Light Source - LCS-0365-76-22	365 nm	300
Second light source	Light Source - LCS-6500-65-22	white	200

Used beam combiner [Name or None]	Beam Combiner - LCS-BC25-0409
Irradiation distance [cm]	9
Thermostat temperature [°C]	25
Irradiation duration [h]	3

Procedure/observations

All steps were carried out using [Protocol] Schlenk Technique. All steps were carried out in the dark, unless other specified.

Date	Time	Step	Observations
29.10		A 10 mL flask was prepared	
	9:50	To the flask [Ru] was added under air	
		The flask was set under argon	
	10:05	To the flask water was added using a 1 mL syringe	
	10:06	To the flask KOH (5 mM) was added using a 50 µL Hamilton syringe	
	10:07 - 10:20	The obtained mixture was stirred at 320 rpm.	before filt.jpg
	10:21	From the obtained solution an NMR sample was prepared according to Protocol - Preparation of NMR Sample using a D2O/H3PO4 capillary. (AE-371-1) During the transfer the sample was filtered through a syringe filter (PA, pore diameter = 0.2 µm)	AE-371-1.jpg

04.11	13:15	The power of the LEDs 365 nm: 300 mW and white: 200 mW were adjusted according to [Protocol] Irradiation setup	power setup .jpg
		The NMR-tube was placed in the setup according to [Protocol] Irradiation setup	irrad setup.jpg
	13:20 - 16:20	The J.Young NMR tube was irradiated for 3:00 h	
	16:50	An NMR was measured (AE-371-2)	AE-371-2.jpg

Analysis

Date	Time	Sample name	Analysis Method	Used Device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
30.11	12:31	AE-371-1	31P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-371-1.zip	AE-371-1_101.nmrium	/	As expected
05.11	15:41	AE-371-2	31P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-371-2.zip	AE-371-1_101.nmrium	/	As expected

Results

Sample	Yield (hydride species) [%]	Yield (dihydroxo) [%]	Data
AE-371-2	3	93	AE-371-2_Yield_Determination.xlsx

comparison with other experiments: [Two-photon - AE-357: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\] \(AE-310\) 365 nm and white, 1000 mM KOH](#)

[Two-photon - AE-358: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\] \(AE-310\) 365 nm and white, 50 mM KOH](#)

[Two-photon - AE-362: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\] \(AE-310\) 365 nm and white, 5 mM KOH](#)

[Two-photon - AE-363: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\] \(AE-310\) 365 nm and white, 1 mM KOH](#)

See excel file: [Comparison-KOH-screening.xlsx](#) and plot: [Figure_1a.png](#) (used for plotting: [Plotting](#))

Faster reaction with more KOH, yielding higher yields, since full conversion is not reached. When yield of hydride normalized to conversion: Data at 0.1 mM difficult to take into account, since very low conversion. Otherwise: Effect less pronounced, but still visible.

At high c(KOH): other side products form instead of the main side product

Linked experiments

- AEI-022: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNN})]$, NO₂-route, 365 nm and white LED
- AEI-026: Capillary production
- AEI-064: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag₂O
- AEI-067: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, crude, 365 nm and white
- AEI-071: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purer, 365 nm and white
- AE-105: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag₂O
- AE-110: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purified, 365 nm and white
- AE-119: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purified, 52 mM KOH, 365 nm and white
- AE-140: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purified, 365 nm and white, longer duration I
- AE-145: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag₂O
- AE-158: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purified, 365 nm and white
- KRA-092: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$, 365 nm and white
- KRA-104: Degassing of Milli-Q water
- AE-238: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag₂O, 5 h
- AE-243: 1 M Stocksolution of KOH in water
- AE-246: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-238, 97 % pure) 365 nm and white
- AE-310: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag₂O, 5 h
- Prep work - AE-361: 5 mM Stocksolution of KOH in water
- Two-photon - AE-348: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, water, longer irradiation
- Two-photon - AE-349: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 50 mM KOH, shorter irradiation
- Two-photon - AE-357: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 1000 mM KOH
- Two-photon - AE-358: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 50 mM KOH

Two-photon - [AE-362: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\] \(AE-310\) 365 nm and white, 5 mM KOH](#)

Two-photon - [AE-363: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\] \(AE-310\) 365 nm and white, 1 mM KOH](#)

Linked resources

Beam Combiner - [Beam Combiner LCS-BC25-0409 -1](#)

Light Source - [UHP LED 365 nm-1](#)

Light Source - [UHP LED 6500 K \(white\)-1](#)

Protocol - [Preparation of NMR Sample](#)

Protocol - [Schlenk Technique](#)

Protocol - [Irradiation setup](#)

Attached files

Comparison-KOH-screening.xlsx

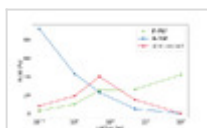
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Plotting

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Figure_1a.png

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AE-371-2_Yield_Determination.xlsx

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AE-371-1_101.nmrium

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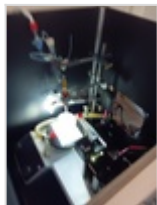
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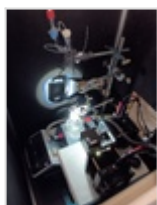
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power-setup-.jpg

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AE-371-1.jpg

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before-filt.jpg

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Unique eLabID: 20241030-b2a33a937ebe5b6408208939291481664ab8795f
Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1459>